

Configuration

Version: 1.0

Purpose

This document defines the configuration architecture of Trading Platform Pro. The configuration system provides a centralized, predictable and environment-aware mechanism for managing application settings while keeping configuration separate from application logic.

Objectives

The configuration system shall provide:

- centralized configuration management
- environment-specific settings
- deterministic loading
- validation during startup
- strong typing
- secure handling of sensitive information

Design Principles

Configuration shall be:

- externalized
- explicit
- validated
- version controlled where appropriate
- independent of business logic

Configuration must never contain business rules.

Configuration Sources

Supported configuration sources include:

- YAML files
- Environment variables
- Command-line arguments
- Runtime overrides (where supported)

The precedence order should be clearly defined and documented.

Environment Profiles

Typical environments:

- development
- testing
- staging
- production

Each environment should contain only the values that differ from the defaults.

Configuration Loading

The startup sequence should:

1. Detect the active profile.
2. Load base configuration.
3. Load environment-specific configuration.

4. Apply environment variables.
 5. Validate the final configuration.
 6. Register configuration in the dependency container.
- Configuration loading should fail fast if validation fails.

Validation

Every configuration value should be validated before runtime.

Typical validation includes:

- required values
- valid paths
- numeric ranges
- enum values
- external resource availability (where practical)

Secrets

Sensitive information must never be stored directly in source code.

Examples:

- API keys
- access tokens
- passwords
- broker credentials

Secrets should be provided through secure external mechanisms.

Configuration Structure

Configuration should be organized by responsibility.

Typical sections:

- Application
- Runtime
- Logging
- Database
- Messaging
- Scheduler
- Plugins
- Broker
- Monitoring

Best Practices

Configuration should be:

- minimal
- readable
- well documented
- backward compatible where practical

Unused configuration entries should be removed.

Configuration Governance

Configuration changes shall:

- be reviewed
- remain backward compatible where practical
- be documented
- be validated before deployment

Configuration is part of the system architecture.

Configuration Review Checklist

Before releasing verify:

- configuration validated
- defaults defined
- secrets externalized
- environment profiles complete
- documentation updated

Related Documents

- Runtime.md
- Architecture.md
- Infrastructure.md
- AGENTS.md